1. Definition of signal

   A **signal** is defined as a variable, a physical variable, or a **function of a physical quantity that conveys information**. For example, temperature can be considered a signal as it conveys information about comfortability.

2. Explain some common signal forms.

Common Signal Forms (from your notes)

1. **Electrical Signal**
   Information represented by **voltage or current**.
   Example: computer circuits, sensors.
2. **Mechanical Signal**
   Information represented by **motion, force, or displacement**.
   Example: vibration in machines.
3. **Optical Signal**
   Information carried by **light waves**.
   Example: fiber-optic communication.
4. **Sound (Acoustic) Signal**
   Information carried through **sound waves** in air.
   Example: human speech (≈ 20 Hz – 20 kHz).
5. **Thermal Signal**
   Information represented by **temperature changes**.
6. **Biological Signal**
   Signals produced by the human body.
   Examples:
   - **ECG** – heart signal
   - **EEG** – brain signal
   - **EMG** – muscle signal
   - **PPG** – blood flow signal

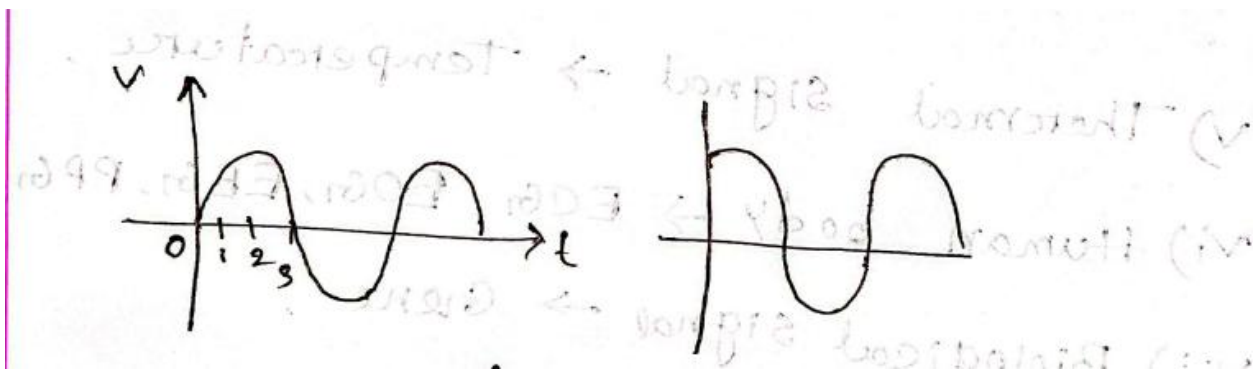
3. def of range, domain.

Domain:
The **domain** of a function is the **set of all possible input values** (independent variable) for which the function is defined.

Range:
The **range** of a function is the **set of all possible output values** that the function can produce.

4.Def of Continuous Time Signal and Discrete Time Signal.

CT Signal: A signal is said to be continuous if it is well defined for all instant of time.



DT Signal: If the signal is well defined for only certain instant of time, then it is called Discrete Time Signal.

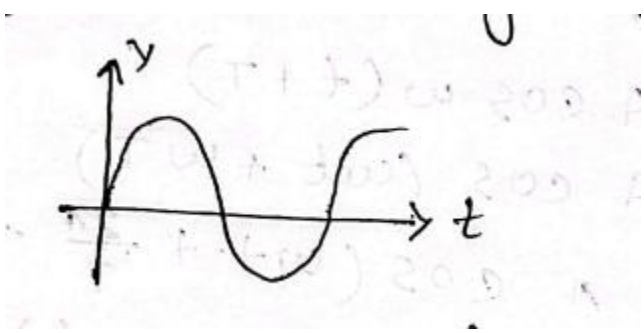
5.Difference between Continuous Time Signal and Discrete Time Signal.

Difference Between Continuous-Time Signal and Discrete-Time Signal

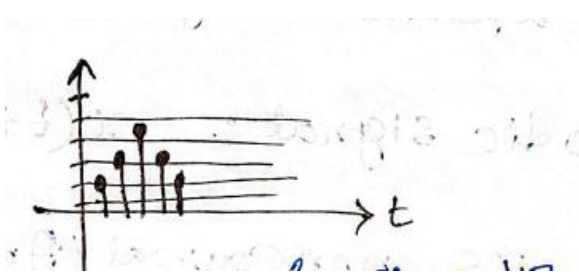
Basis	**Continuous-Time Signal**	**Discrete-Time Signal**
Definition	Defined at **every instant of time**	Defined only at **specific time intervals**
Representation	$x(t)$	$x[n]$
Time variable	**Continuous**	**Discrete (integer values)**
Nature	Usually **analog**	Usually **digital**
Example	Microphone audio signal	Digital audio samples in a computer

6. Def of Analog Signal and Digital Signal.

Analog: If both axis are continuous time in nature then it is analog Signal

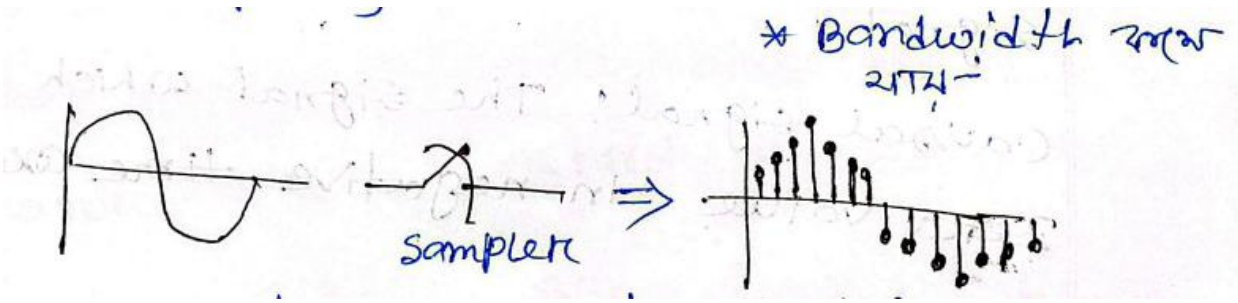


Digital: If both axis are discrete time in nature then it is digital Signal

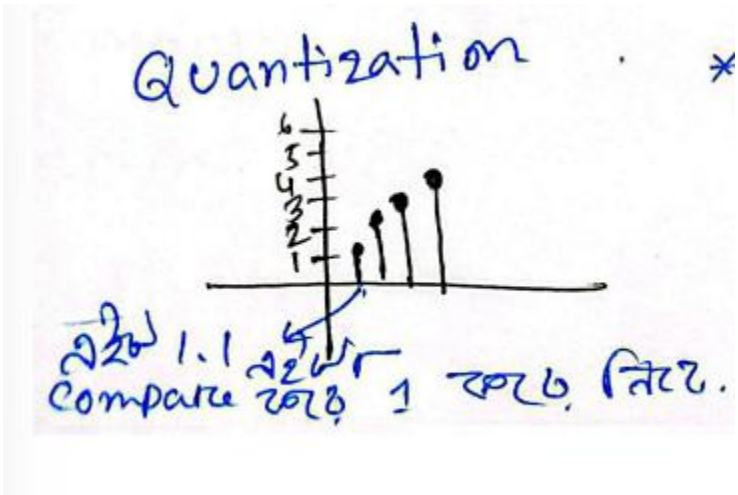


7.Def of sampling and quantization.

Sampling: Sampling is the process of **converting a continuous-time signal into a discrete-time signal by taking values at regular time intervals.**



Quantization:
Quantization is the process of **converting the continuous amplitude values of a sampled signal into a finite set of discrete levels.**



8. Difference between sampling and quantization.

Here's a concise table version:

Feature	Sampling	Quantization
Purpose	Discretize **time**	Discretize **amplitude**
Input	Continuous-time signal	Sampled signal
Output	Discrete-time signal	Discrete-time, discrete-amplitude signal
Error	Possible aliasing if under-sampled	Quantization error (rounding)

9. Def of periodic and non-periodic signal

Periodic: A signal that **repeats itself exactly after a fixed interval** T. Example: $x(t) = sin(t)$.

Ex. $x(t) = A \cos \omega t$

$$x(t+T) = A \cos \omega (t+T)$$

$$= A \cos (\omega t + \omega T)$$

$$= A \cos \left(\omega t + \frac{2\pi}{T} t\right) \quad \left[\omega = \frac{2\pi}{T}\right.$$

$$= A \cos (2\pi + \omega t)$$

$$A \cos \omega (t+T) = A \cos \omega t$$

This is a perciodic signal.

Non-Periodic: A signal that **does not repeat itself** over time.
Example: a single pulse or random signal.

$$x(t+T) \neq x(t)$$

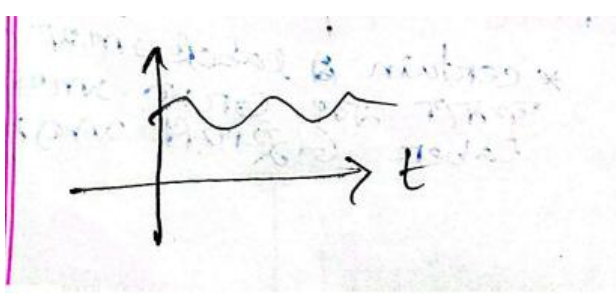
10. Difference between periodic and non-periodic signal

Here's a clear table showing the difference:

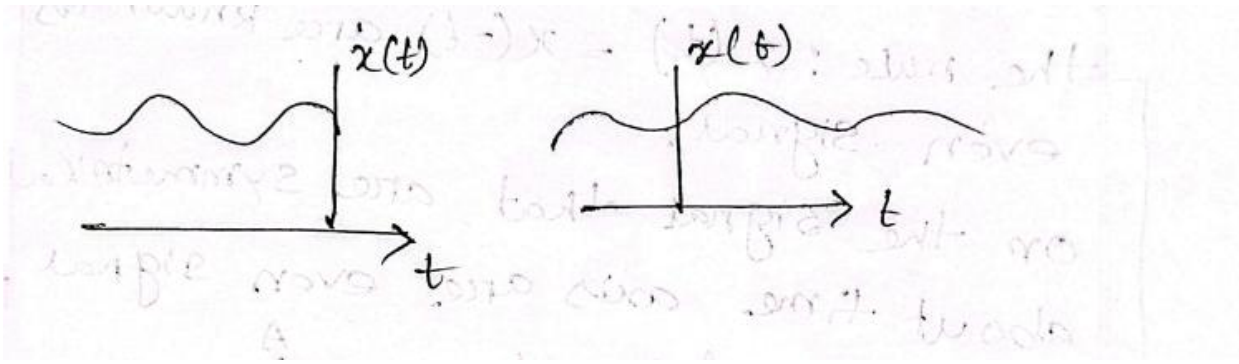
Feature	**Periodic Signal**	**Non-Periodic Signal**
Definition	Repeats itself after a fixed time period T	Does not repeat over time
Time Period	Has a well-defined period T	No fixed period
Example	$x(t) = \sin(t), \cos(t)$	Single pulse, random noise
Frequency Spectrum	Discrete (line spectrum)	Continuous spectrum
Predictability	Predictable	Unpredictable

11. Def of casual and anti-casual signal

Casual: The signal which have zero value in negative time axis.



Anti-Causal: The signal which have zero value in positive time axis or value exists in both axis.



12. Difference between casual and anti-casual signal.

Here's a clear comparison in table form:

Feature	**Causal Signal**	**Anti-Causal Signal**
Definition	Zero for all $t < 0$; exists from present/future	Zero for all $t > 0$; exists only in past
Time Support	$t \geq 0$	$t \leq 0$
Example	$x(t) = e^{-t}u(t)$	$x(t) = e^{t}u(-t)$
Physical Realizability	Often physically realizable	Usually theoretical, not physically realizable

13. Def of Deterministic and Random signal.

Deterministic: The signal which obeys mathematical rules and can be expressed by mathematical equations are called deterministic signal.

Random: The signal which doesnot obeys mathematical rules and random in nature are called random signal. Also known as noise signal.(Unwanted signal that interrupt main signal)

14. Difference between Deterministic and Random signal.

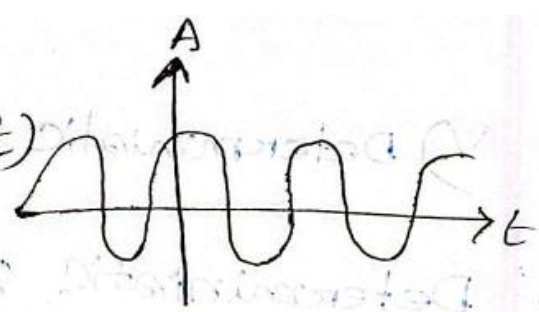
Here's a concise table comparing **Deterministic** and **Random signals**:

Feature	**Deterministic Signal**	**Random (Stochastic) Signal**
Definition	Can be **exactly predicted** at any time	Cannot be precisely predicted; **random behavior**
Predictability	Fully predictable	Not predictable
Example	$x(t) = \sin(t), x[n] = 2^n$	Noise, stock prices, thermal signal
Mathematical Model	Exact formula exists	Described statistically (mean, variance)
Reproducibility	Reproducible	Not exactly reproducible

15. Def of Even and Odd signal.

Even: The signal that obeys the rule: x(t) = x(-t) are known as even signal. Or, The signal that are symmetrical about time axis are even signal.

$$x(t) = A\cos\omega t$$
$$x(-t) = A\cos\omega(-t)$$
$$= A\cos\omega t$$
$$x(t) = x(-t)$$



Odd: The signals which are asymmetrical about time axis are called odd signal.

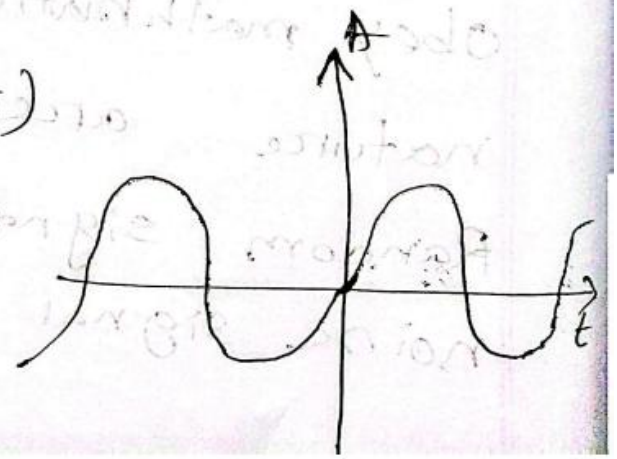
Mathmatically, $x(t) = -x(-t)$

$x(t) = A\sin \omega t$

$x(-t) = A \sin \omega(-t)$

$x(-t) = -A \sin \omega t$

$x(-t) = -x(t)$

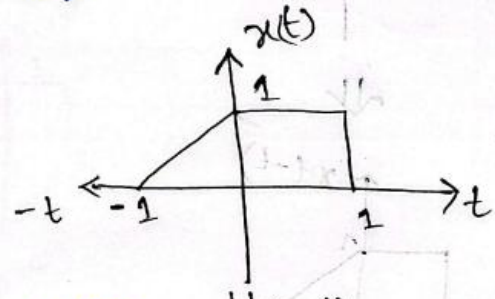


even, $x_e(t) = \dfrac{x(t)+x(-t)}{2}$

odd, $x_o(t) = \dfrac{x(t)-x(-t)}{2}$

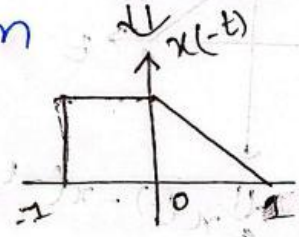
Math 1:-

Plot even & odd signal from the given signa



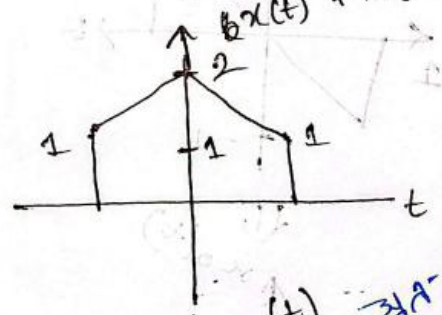
⇒ Soln: even

Just shift হবে -

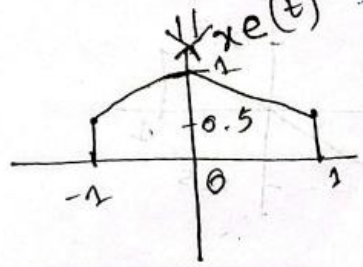


→ formula even

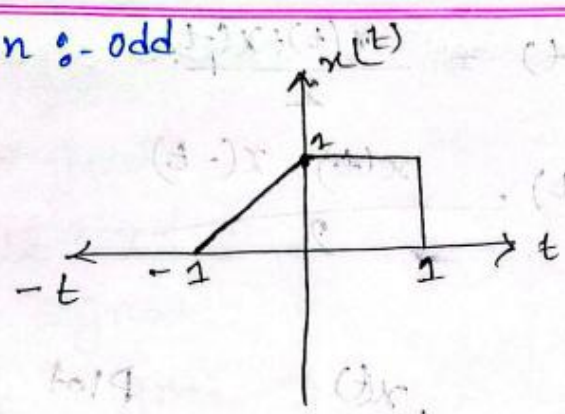
$x(t) + x(-t)$

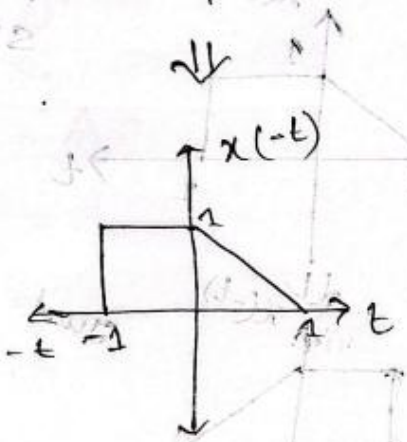


সুতরাং 2 দ্বারা ভাগ করলে half হবে।

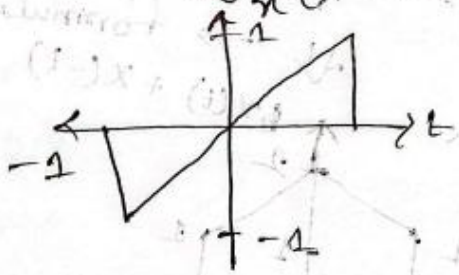


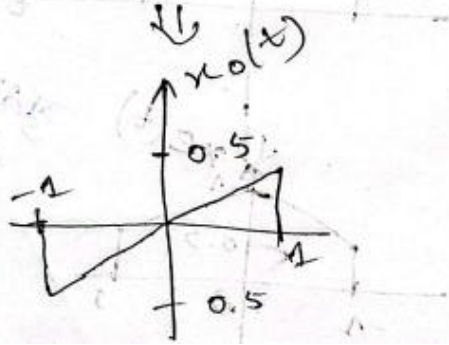
⇒ soln :- odd



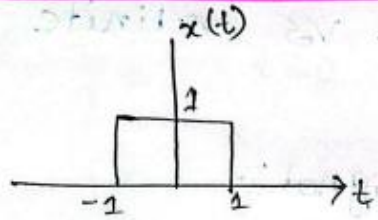


$x(t) - x(-t)$ Formula



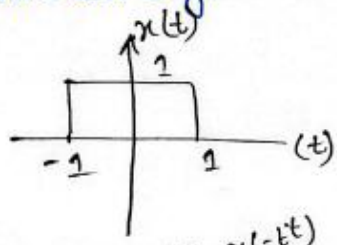


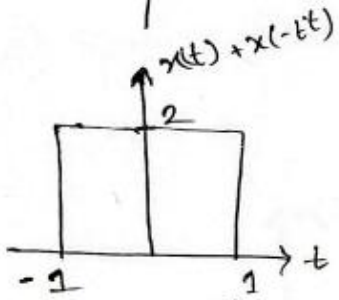
Q.2 :-

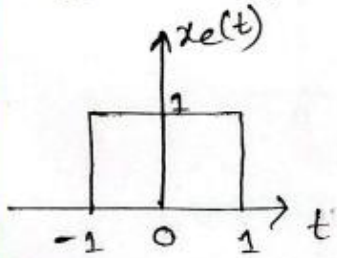


Plot the graph of even and odd signal using the given signal.

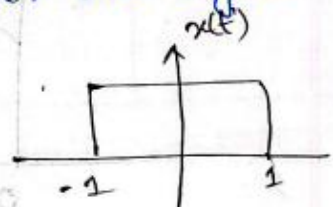
For even signal :-

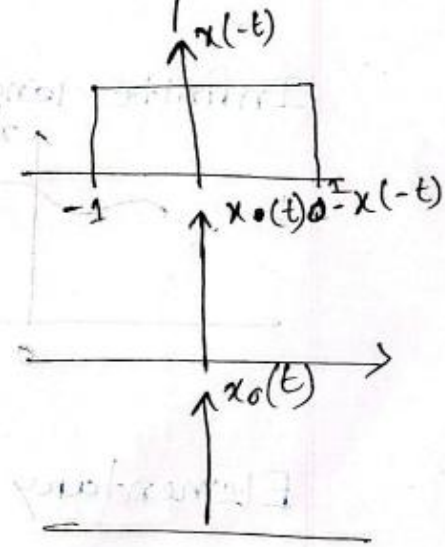






For odd signal.





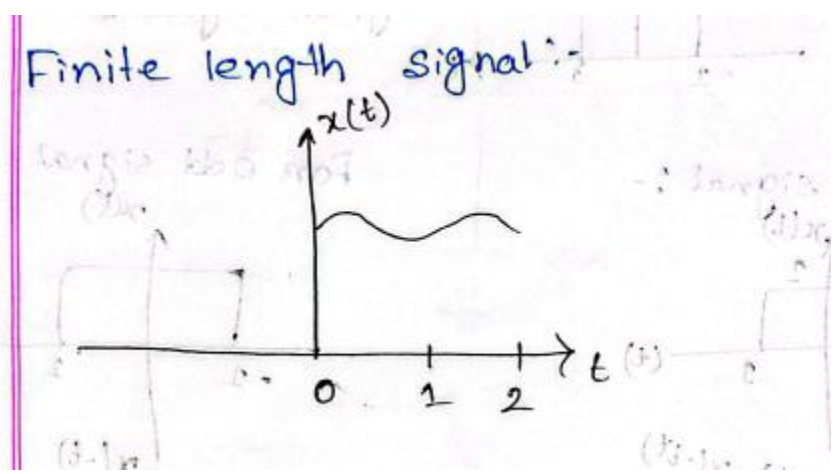
16. Difference between Even and Odd signal.

Here's a clear table comparing **Even** and **Odd signals**:

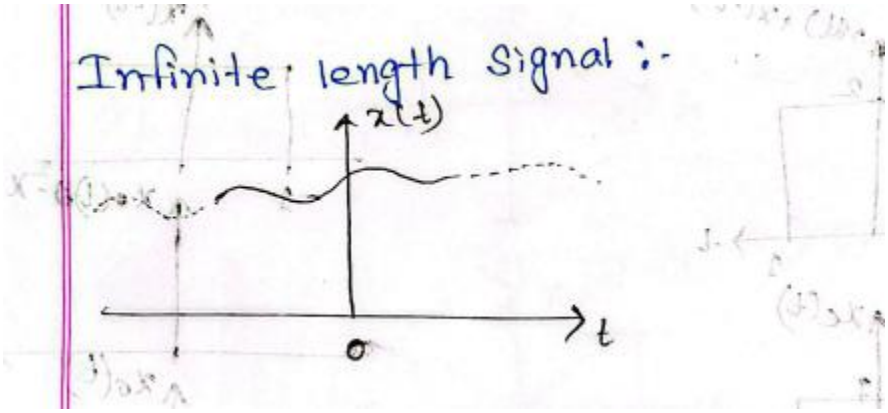
Feature	**Even Signal**	**Odd Signal**
Definition	Symmetric about y-axis: $x(t) = x(-t)$	Anti-symmetric: $x(t) = -x(-t)$
Graph Symmetry	Mirror image about y-axis	Rotational symmetry about origin
Example	$x(t) = t^2, \cos(t)$	$x(t) = t, \sin(t)$
Sum of Values	Not necessarily zero	Sum over symmetric interval is zero
Decomposition	Any signal can be expressed as sum of **even + odd**	Same (part of decomposition)

17.Def of Finite length and infinite length signal.

Finite Length: A signal that is **non-zero only for a finite duration** of time. Example: a pulse signal lasting from $t = 0$to $t = 5$.



Infinite Length: A signal that **exists for all time** (never becomes zero completely). Example: $x(t) = \sin(t)$or $x[n] = 0.9^n$for $n \geq 0$.



18. Difference between Finite Length and infinite Length signal.

Here's a clear comparison in table form:

Feature	**Finite Length Signal**	**Infinite Length Signal**
Definition	Non-zero for a **limited duration** of time	Exists for **all time** (never fully zero)
Time Support	Limited (e.g., $t_1 \leq t \leq t_2$)	Unlimited ($-\infty < t < \infty$)
Example	A pulse from $t = 0$to $t = 5$	$x(t) = \sin(t), x[n] = 0.9^n$
Analysis Complexity	Easier to analyze mathematically	More complex, may need special techniques